HCF-Net: Hierarchical Context Gating for Neonatal Brain Lesion Segmentation

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ABSTRACT

Accurate segmentation of hypoxic ischemic encephalopathy (HIE) lesions in neonatal brain MRI is critical for clinical assessment but remains challenging due to the lesions' highly variable morphology, scale, and contrast. We propose HCF-Net, a novel segmentation architecture built upon a hierarchical context gating framework designed to address these challenges. The core innovation is the replacement of standard skip connections with Cross-Frequency Gating Blocks, which enable adaptive feature transmission between encoder and decoder levels by operating on frequency-decomposed representations. This allows low-frequency components to guide global context selection while high-frequency components preserve essential boundary details. The encoder and decoder pathways are constructed using parameter-efficient Multi-Scale Dilated Separable Convolution blocks to capture contextual information at multiple scales without spatial downsampling. The entire framework is designed with linear-complexity operations compatible with 3D volumetric constraints. Validated on the publicly available BONBID2023 dataset of neonatal brain diffusion MRI, HCF-Net shows effective segmentation of variable-scale HIE lesions, balancing computational efficiency with adaptive, multi-scale feature integration. The code and checkpoints are available at https://github.com/yifangao112/HCF-Net

1. Introduction

Hypoxic ischemic encephalopathy (HIE) is a leading cause of neonatal mortality and long-term neurological disability, resulting from a critical reduction in cerebral blood flow and oxygen supply around the time of birth. Accurate identification and volumetric assessment of HIE-related brain lesions from magnetic resonance imaging (MRI) are crucial for clinical decision-making, including identifying high-risk neonates, evaluating therapeutic interventions like therapeutic hypothermia, and predicting long-term neurodevelopmental outcomes Shen, Xia, Kang, Yuan, and Sheng (2011). While diffusion-weighted MRI and its derived Apparent Diffusion Coefficient (ADC) maps are sensitive to the cytotoxic edema characteristic of acute HIE, manual segmentation of these lesions is a time-consuming, subjective process that requires specialized neuroradiological expertise Toubal, Kazemi, Rahmon, Kucukpinar, Almansour, Ho, and Palaniappan (2023). Consequently, there is a significant clinical need for robust, automated segmentation tools to support quantitative analysis and improve diagnostic consistency.

The development of such automated tools, however, is confronted by several inherent challenges specific to neonatal HIE lesion morphology. Lesions exhibit highly variable appearance in terms of scale, shape, and contrast, ranging from small, punctate foci to large, confluent regions of injury Weiss, Bates, Song, Zhang, Herzberg, Chen, Gong, Chien, Zhang, and Murphy (2019). Over half of the patients present with lesions occupying less than 1% of the total brain volume, making them particularly difficult to detect against complex anatomical backgrounds Toubal et al. (2023). Effective segmentation therefore necessitates the adaptive integration of multi-scale contextual information to simultaneously capture fine-grained lesion boundaries for precise localization and global anatomical context for accurate tissue classification. While encoder-decoder architectures like U-Net Ronneberger, Fischer, and Brox (2015) and its numerous successors Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018) have become the de facto standard for medical image segmentation, their standard skip connections employ fixed,

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additive fusion of encoder and decoder features. This design is suboptimal for HIE, as it cannot adaptively weight the contribution of multi-scale features based on the specific characteristics of a variable-scale lesion.

Recent advances have sought to address these limitations through more sophisticated feature fusion mechanisms. Transformer-based models employ global self-attention to model long-range dependencies, but their quadratic computational complexity is often prohibitive for high-resolution 3D medical volumes. Efficiency-oriented adaptations using localized window attention Liu, Lin, Cao, Hu, Wei, Zhang, Lin, and Guo (2021) or state-space models Gu and Dao (2023) reduce computational cost but may sacrifice the cross-scale contextual awareness needed for variable lesion sizes. Alternative approaches operating in the frequency domain Chen, Tu, Liu, Li, and Plaza (2025) offer linear complexity by compressing features via spectral decomposition; however, a common trade-off is the discarding of high-frequency components that are essential for preserving detailed boundary information. Consequently, a critical gap remains: a segmentation architecture that balances computational efficiency with adaptive, multi-scale feature integration, explicitly preserving both global context and local detail, all within the practical constraints of 3D medical image analysis.

In this paper, we propose HCF-Net, a novel Hierarchical Context Fusion Network designed specifically for the challenging task of neonatal HIE lesion segmentation. Our core innovation is a linear-complexity Cross-Frequency Gating (CFG) block that replaces standard skip connections. The CFG block decomposes encoder and decoder features into frequency-domain representations, using compressed low-frequency components to compute adaptive gating weights that guide global context integration, while explicitly preserving a residual high-frequency pathway to maintain boundary details. This gating mechanism is embedded within a U-Net-like structure where the encoder and decoder are themselves constructed using parameter-efficient Multi-Scale Dilated Separable Convolution (MSDSC) blocks. These blocks capture contextual information at multiple scales through parallel dilated convolutions without spatial downsampling, making them well-suited for processing the constrained 3D patch sizes typical in medical imaging pipelines.

The principal contributions of this work are as follows. We introduce the Cross-Frequency Gating block, a novel attention mechanism that performs adaptive hierarchical feature fusion in the frequency domain with linear complexity. It uniquely preserves high-frequency components for boundary definition while using low-frequency context for adaptive gating. We design a parameter-efficient Multi-Scale Dilated Separable Convolution block that captures multi-scale contextual information while maintaining spatial resolution, serving as the fundamental feature extractor throughout the network. We integrate these components into HCF-Net and show its effectiveness on the publicly available BONBID2023 HIE segmentation dataset. Our method achieves state-of-the-art performance, with a Dice score of 0.5823 and an HD95 of 11.9295, outperforming strong contemporary baselines including MedNeXt, nnUNet, and UNETR. Comprehensive ablation studies confirm the contribution of each proposed component.

The remainder of this paper is organized as follows. We review related work in Section 2. The detailed methodology of HCF-Net is presented in Section 3. Experimental setup, results, and analyses are provided in Section 4. Finally, we conclude with a discussion of limitations and future directions in Section 5.

2. Related Work

2.1. Medical Image Segmentation Architectures

Convolutional encoder-decoder architectures, particularly the U-Net family, have established the foundational paradigm for medical image segmentation. The original U-Net Ronneberger et al. (2015) introduced a symmetric encoder-decoder structure with skip connections to combine high-resolution encoder features with semantically rich decoder features, enabling precise localization. This design has inspired numerous successful variants, including the 3D V-Net Milletarì, Navab, and Ahmadi (2016) for volumetric data, the Attention U-Net which adds spatial attention gates to skip connections, and U-Net++ Zhou, Siddiquee, Tajbakhsh, and Liang (2018) with nested, dense skip pathways. The highly effective nnU-Net Isensee, Petersen, Klein, Zimmerer, Jaeger, Kohl, Wasserthal, Koehler, Norajitra, and Wirkert (2018) framework further showed that robust, problem-agnostic training pipelines combined with this architectural template can achieve state-of-the-art performance across diverse challenges. However, a core limitation persists: the feature fusion in these models relies on static, concatenation-based skip connections. These connections transmit features with fixed, non-adaptive weighting, which is suboptimal for segmenting lesions

with highly variable morphology and scale, as they cannot dynamically prioritize the most relevant contextual information from the encoder for a given decoder state.

To capture long-range dependencies and global context beyond the local receptive fields of convolutions, transformer-based segmentation models have been extensively explored. Architectures like UNETR Hatamizadeh, Yang, Roth, and Xu (2021) and Swin UNETR Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022) replace the convolutional encoder with a Vision Transformer (ViT) or its hierarchical Swin Transformer variant, using a convolutional decoder to reconstruct high-resolution segmentations. Other approaches, such as TransUNet Chen, Lu, Yu, Luo, Adeli, Wang, Lu, Yuille, and Zhou (2021) and CoTr Xie, Zhang, Shen, and Xia (2021), integrate transformers within the bottleneck or as bridging modules between convolutional encoders and decoders. While these models excel at modeling global contextual relationships, their computational complexity, which scales quadratically with the number of input tokens, presents a significant bottleneck for high-resolution 3D medical volumes. Efficiency-oriented adaptations, such as the windowed attention in Swin Transformers Liu et al. (2021) or axial attention Yan, Tang, Sun, Ma, Kong, and Xie (2021), reduce this burden but often at the cost of limiting cross-window or cross-scale contextual awareness, which can be detrimental for capturing the full extent of variable-scale pathological structures.

A parallel line of research focuses on designing efficient modules for multi-scale feature extraction and fusion within convolutional frameworks, seeking a better trade-off between contextual modeling and computational cost. Dilated convolutions Yu and Koltun (2015) and atrous spatial pyramid pooling (ASPP) Zhu, Chen, Hu, Fu, Xu, Qin, and Heng (2020) are widely adopted to expand receptive fields without downsampling. More recent innovations explore frequency-domain processing for efficient global modeling; for instance, methods like FNet Lee-Thorp, Ainslie, Eckstein, and Ontañón (2021) and GFNet Rao, Zhao, Zhu, Lu, and Zhou (2021) use Fast Fourier Transforms (FFT) for token mixing. However, many frequency-domain approaches apply aggressive low-pass filtering or discard high-frequency components to achieve efficiency, which can erase fine-grained boundary details critical for accurate segmentation. Similarly, while local window attention mechanisms are efficient, they lack explicit mechanisms for adaptive, hierarchical fusion across different feature scales. This landscape reveals a distinct gap: there is a need for segmentation architectures that can perform adaptive, content-aware fusion of multi-scale features with linear computational complexity, while explicitly preserving both low-frequency contextual information and high-frequency boundary details, particularly within the constraints of 3D volumetric data processing.

2.2. Efficient Context Modeling and Feature Fusion

Efficient attention mechanisms have been developed to mitigate the quadratic complexity of standard self-attention, which is prohibitive for 3D medical volumes. Linear attention approaches approximate the softmax operation using kernel functions or low-rank decompositions to achieve linear scaling with sequence length Katharopoulos, Vyas, Pappas, and Fleuret (2020) Katharopoulos et al. (2020) Wang, Li, Khabsa, Fang, and Ma (2020). While effective for reducing computational burden, these approximations often trade off the ability to capture complex, long-range dependencies, which are crucial for understanding global anatomical context in brain MRI. Alternatively, local window attention restricts the receptive field to a fixed neighborhood, significantly lowering memory consumption Liu et al. (2021) Hatamizadeh et al. (2022) Shaker, Maaz, Rasheed, Khan, Yang, and Khan (2022). However, this design inherently limits cross-scale contextual awareness, as features from distant but semantically relevant regions cannot interact directly, a critical shortcoming for segmenting diffuse lesions that may span multiple anatomical regions.

Frequency-domain methods offer another pathway for efficient global context modeling by transforming spatial features into a compact spectral representation. Techniques based on the Discrete Cosine Transform (DCT) or Fast Fourier Transform (FFT) compress feature maps by retaining only low-frequency components, enabling efficient global operations Rao et al. (2021). This compression is particularly appealing for 3D data, as it reduces the dimensionality of the feature space for subsequent processing. A significant drawback of this strategy is the systematic discarding of high-frequency information, which corresponds to fine details such as edges and texture. For medical image segmentation, where precise boundary delineation of small lesions is paramount, this loss of high-frequency signal can lead to blurred predictions and poor performance on challenging, low-contrast boundaries Imtiaz, Fattah, and Kung (2023) Jiao, Tang, Lin, Gao, Ma, Wang, and Zheng (2023).

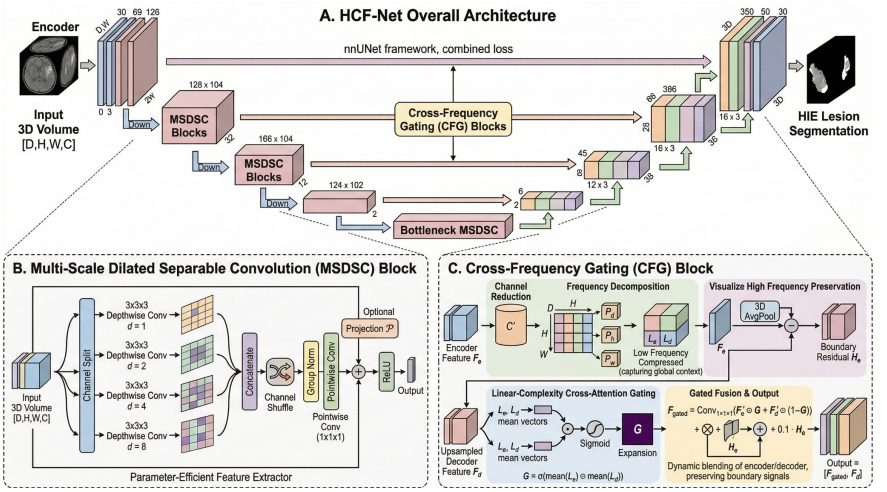

Figure 1: Overall architecture of HCF-Net. The hierarchical encoder-decoder network replaces standard U-Net components with Multi-Scale Dilated Separable Convolution (MSDSC) blocks for feature extraction and Cross-Frequency Gating (CFG) blocks for adaptive hierarchical fusion, enabling efficient multi-scale context integration for 3D HIE lesion segmentation.

Within convolutional frameworks, specialized designs aim to expand receptive fields and improve parameter efficiency for multi-scale context capture. Dilated (or atrous) convolutions systematically insert zeros into convolutional kernels to exponentially increase the receptive field without increasing parameters or losing resolution, forming the basis of architectures like DeepLab Chen, Zhu, Papandreou, Schroff, and Adam (2018) and its 3D medical adaptations Chen et al. (2018). While effective, stacking dilated convolutions can lead to gridding artifacts and a loss of local continuity. Separable convolutions, including depthwise and pointwise variants, dramatically reduce computational cost by factorizing the standard convolution operation Sandler, Howard, Zhu, Zhmoginov, and Chen (2018) Chollet (2016). These have been successfully integrated into medical segmentation networks like UNet++ and nnU-Net variants to maintain efficiency Zhou et al. (2018). However, most existing designs either focus on receptive field expansion or parameter reduction in isolation, lacking a unified mechanism for efficient, artifact-free multi-scale feature extraction that is both computationally lightweight and preserves fine spatial details essential for 3D volumetric analysis.

3. Method

3.1. Overall Architecture

The HCF-Net architecture is designed as a hierarchical encoder-decoder network that addresses the specific challenges of segmenting HIE lesions, which exhibit highly variable morphology and scale within constrained 3D volumetric data. The framework is built upon a modified U-Net structure where standard components are systematically replaced to enable efficient, adaptive multi-scale feature integration. The core design philosophy is to maintain linear computational complexity relative to input size while facilitating context-aware fusion across hierarchical levels, thereby balancing the capture of global anatomical context with the preservation of fine-grained boundary details.

The encoder and decoder pathways are constructed using stacks of Multi-Scale Dilated Separable Convolution (MSDSC) blocks, which serve as the fundamental feature extraction units. This design replaces the standard 3×3×3 convolutions found in conventional U-Nets. Each MSDSC block captures contextual information at multiple scales through parallel dilated convolutions without resorting to spatial downsampling, a critical feature given the small input patch dimensions typical for 3D medical image analysis. The output from each encoder level is subsequently processed not by a direct skip connection, but by a Cross-Frequency Gating (CFG) block. This substitution represents the central innovation of HCF-Net, transforming the feature transmission pathway from a static concatenation into an adaptive, attention-guided gating mechanism.

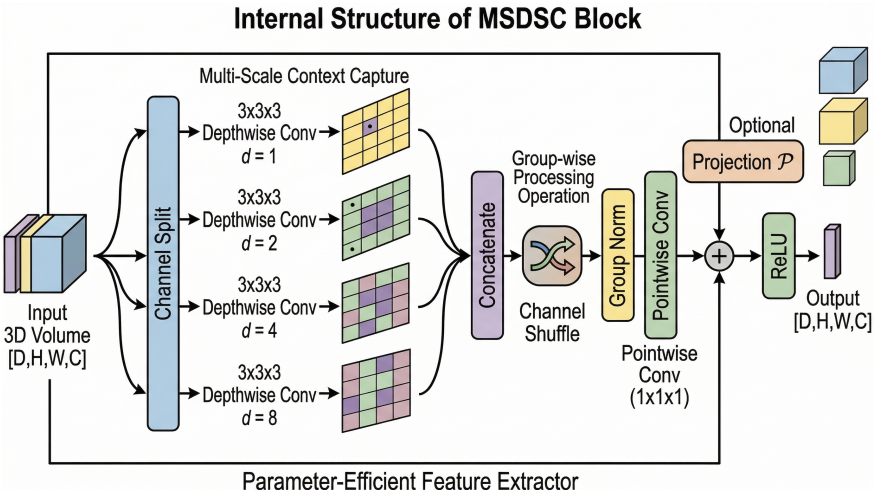

Figure 2: Structure of the MSDSC block. The input is split into four channel groups, each processed by a depthwise convolution with an exponentially increasing dilation rate (d=1,2,4,8). Outputs are concatenated, shuffled, fused via pointwise convolution, and combined with a residual projection to capture multi-scale context without spatial downsampling.

The integrated data flow through HCF-Net creates a hierarchical context fusion system. Features extracted by the encoder at a given level are decomposed into frequency components within the corresponding CFG block. Concurrently, features from the decoder at the same hierarchical level undergo a similar decomposition. A lightweight cross-attention operation is then computed between the compressed low-frequency representations of these two feature streams. The resulting gating weights adaptively modulate the encoder features before they are fused with the decoder pathway, ensuring that the most relevant contextual information is propagated based on the current decoding state. This process is repeated at each encoder-decoder connection, establishing a gated hierarchy that dynamically controls information flow across scales. This integrated design, combining MSDSC blocks for multi-scale feature extraction and CFG blocks for adaptive hierarchical gating, results in a parameter-efficient model capable of addressing the complex segmentation task of HIE lesions in neonatal brain MRI.

3.2. Multi-Scale Dilated Separable Convolution Block

The Multi-Scale Dilated Separable Convolution (MSDSC) block is designed as a parameter-efficient and computationally frugal feature extractor that captures contextual information across multiple scales without resorting to spatial downsampling. This design is critical for segmenting HIE lesions from small volumetric patches, where aggressive downsampling would discard the fine-grained details of small lesions, yet a large receptive field is required to understand their anatomical context. The block addresses this by employing parallel dilated convolutions within a depthwise separable framework, enabling the network to perceive patterns at exponentially increasing spatial scales while maintaining a linear computational complexity with respect to the input size, a prerequisite for processing 3D medical data Chen, Papandreou, Schroff, and Adam (2017) Tang, Jiang, Yao, and Wu (2025).

The core operation of the MSDSC block begins by splitting the input feature map along the channel dimension into four equal groups. Each group is then processed by an independent depthwise convolution with a distinct dilation rate. Formally, for an input feature tensor $X \in \mathbb{R}^{C\times D\times H\times W}$, the channel splitting operation yields four subtensors:

$$\{X_1, X_2, X_3, X_4\} = \text{split}(X), \quad \text{where } X_i \in \mathbb{R}^{(C/4)\times D\times H\times W}. \tag{1}$$

Each subtensor X_i is convolved with a $3\times 3\times 3$ depthwise kernel K_i^{dw}, applying a dilation rate $d = 2^{i-1}$ to exponentially expand the receptive field. The depthwise convolution for the i-th branch is defined as:

$$Y_i(c, \mathbf{p}) = \sum_{\delta\in\Delta} K_i^{\text{dw}}(c, \delta) \cdot X_i(c, \mathbf{p} + d \cdot \delta), \tag{2}$$

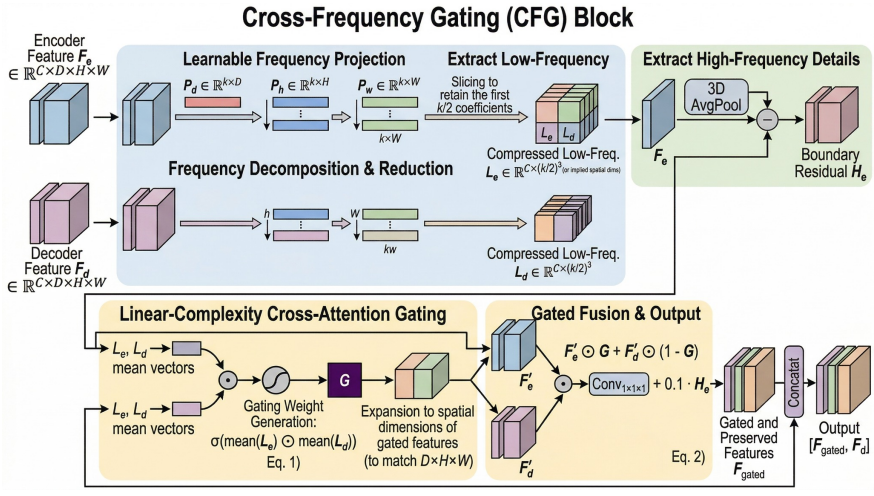

Figure 3: Workflow of the CFG block. Encoder and decoder features are reduced in channels and projected into a frequency representation. Low-frequency components guide a linear-complexity cross-attention gating mechanism, which dynamically modulates encoder features before fusion, while high-frequency residuals are explicitly preserved for boundary details.

where $\mathbf{p} = (d, h, w)$ indexes the spatial position, Δ defines the 3^3 kernel offsets, and c indexes the channel within the group. This parallel processing captures local texture ($d = 1$), intermediate context ($d = 2, 4$), and broader anatomical relationships ($d = 8$) simultaneously.

To facilitate cross-scale interaction among the features extracted at different dilation rates, the outputs Y_i are concatenated and then subjected to a channel shuffle operation. The concatenation produces $Y_{\text{cat}} = [Y_1, Y_2, Y_3, Y_4] \in \mathbb{R}^{C\times D\times H\times W}$. The channel shuffle systematically interleaves the channels from the four groups, promoting information flow across the different scale-specific representations Zhang, Zhou, Lin, and Sun (2017). This is followed by a pointwise convolution that performs cross-channel fusion and projects the features. The complete fusion and projection step, which includes group normalization for stable training, is given by:

$$Z = W_{\text{pw}}\text{GroupNorm}(\text{Shuffle}(Y_{\text{cat}})), \tag{3}$$

where $W_{\text{pw}} \in \mathbb{R}^{C\times C\times 1\times 1\times 1}$ is the pointwise convolution kernel and

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$$\tag{4}$$

Here, $\mathcal{P}$ denotes a projection, implemented as a $1 \times 1 \times 1$ convolution when the input and output channel dimensions differ, or as the identity function otherwise. This residual design ensures that even as multi-scale context is aggregated, the original signal is preserved, which is vital for accurately localizing small lesion boundaries. The MSDSC block thus serves as the fundamental building unit in both the encoder and decoder pathways of HCF-Net, replacing standard convolutions to provide a multi-scale receptive field at every hierarchical level without compromising spatial resolution or computational efficiency.

3.3. Cross-Frequency Gating Block

The Cross-Frequency Gating (CFG) block constitutes the core innovation for adaptive hierarchical fusion within HCF-Net, designed to overcome the limitations of static skip connections in standard encoder-decoder architectures. In the context of segmenting highly variable HIE lesions, where global anatomical context and fine-grained boundary details are both critical, the CFG block enables a dynamic, data-dependent feature selection mechanism. It operates by decomposing features into their frequency-domain representations, allowing low-frequency components to guide the integration of global context while explicitly preserving high-frequency information essential for defining lesion boundaries. This approach provides a computationally efficient alternative to quadratic-complexity attention mechanisms, making it suitable for 3D volumetric data.

The technical formulation begins with channel reduction and frequency decomposition of the input features. Given an encoder feature map $F_e \in \mathbb{R}^{C\times D\times H\times W}$ and its corresponding decoder feature map F_d at the same hierarchical level, we first apply a pointwise convolution to reduce channel dimensionality for computational efficiency, yielding F'_e and F'_d with $C' = \max(C/4, 8)$ channels. The minimum of 8 channels ensures a sufficient feature representation even at deeper network levels. Subsequently, a learnable frequency projection is applied independently along each spatial dimension. Inspired by the Discrete Cosine Transform (DCT) Zhou (2011), this projection uses learnable basis matrices $P_d \in \mathbb{R}^{k\times D}$, $P_h \in \mathbb{R}^{k\times H}$, and $P_w \in \mathbb{R}^{k\times W}$, where $k = 16$ defines the frequency rank. The projection compresses the spatial information into a compact spectral representation. From this, we extract the low-frequency components by retaining the first $k/2$ coefficients along each dimension, forming highly compressed representations $L_e, L_d \in \mathbb{R}^{C'\times(k/2)}$ that capture global context. Concurrently, high-frequency details are explicitly preserved by computing a residual signal from the original encoder features: $H_e = F_e - \text{AvgPool3d}(F_e)$, where a 3D average pooling with a small kernel provides a locally smoothed signal whose subtraction isolates high-frequency boundary information.

The adaptive gating mechanism is computed via a linear-complexity cross-attention operation on the compressed low-frequency representations. This design choice is motivated by the need for efficiency; operating on the drastically reduced L_e and L_d tensors avoids the quadratic cost associated with spatial self-attention. The gating weights G are generated by first computing channel-wise mean vectors from L_e and L_d to obtain a global spectral signature for each feature map. An element-wise product between these signatures followed by a sigmoid activation σ produces a compact gating vector, which is then spatially expanded to match the dimensions of F'_e and F'_d.

$$G = \sigma\left(\text{mean}(L_e) \odot \text{mean}(L_d)\right) \quad \text{(expanded spatially)} \tag{5}$$

This process allows the decoder's current contextual understanding, encoded in L_d, to adaptively gate which encoder features from F'_e are most relevant for the current segmentation task.

The final fusion step combines the gated features with the preserved high-frequency details. The modulated features are computed as a weighted sum: $F'_e \odot G + F'_d \odot (1 - G)$, where the gating map G dynamically blends information from the encoder and decoder streams. This sum is then processed by a pointwise convolution to restore channel interactions, and the explicitly preserved high-frequency residual H_e is added with a small weighting factor of 0.1.

$$F_{\text{gated}} = \text{Conv}_{1\times1\times1}\left(F'_e \odot G + F'_d \odot (1 - G)\right) + 0.1 \cdot H_e \tag{6}$$

The weighting of H_e is kept low to prevent high-frequency noise from dominating the fused features while ensuring boundary signals are retained. The output of the CFG block, which replaces the standard skip

connection, is the concatenation $[F_{\text{gated}}, F_d]$. This output is then passed to the corresponding decoder layer, creating a hierarchical gating mechanism where feature transmission is conditioned on cross-frequency attention, thereby enabling more precise and adaptive multi-scale context fusion throughout the network.

3.4. Hierarchical Integration and Complexity Analysis

The HCF-Net architecture integrates the Multi-Scale Dilated Separable Convolution (MSDSC) and Cross-Frequency Gating (CFG) blocks into a coherent hierarchical framework for adaptive feature fusion. The encoder and decoder pathways are each constructed from sequential stacks of MSDSC blocks, which serve as the fundamental feature extractors, capturing multi-scale contextual information without spatial downsampling. At each corresponding level between the contracting and expanding pathways, the standard skip connection is replaced by a CFG block. This establishes a hierarchical gating mechanism where features from the encoder and decoder are not simply concatenated but are first decomposed into their frequency-domain representations. The data flow through this mechanism is designed to be both adaptive and efficient. For a given hierarchical level, the encoder feature F_e and the upsampled decoder feature F_d are processed by the CFG block. Their low-frequency components are extracted and used to compute a cross-attention gating map, which dynamically modulates the contribution of F_e to the fusion process. The gated encoder features are then concatenated with the decoder stream F_d, allowing the network to selectively emphasize or suppress contextual information from the encoder based on the decoder's current state and the global anatomical context encoded in the low-frequency signals.

A formal analysis of the computational complexity reveals the efficiency gains of this design, particularly for 3D volumetric data. The critical operation is the frequency projection within the CFG block, which replaces the quadratic-complexity global self-attention common in transformer-based models Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, and Polosukhin (2017). Standard self-attention on a feature map of size $C \times N$, where $N = D \times H \times W$ is the number of spatial tokens, has a computational complexity of $\mathcal{O}(N^2)$ due to the pairwise similarity computation. In contrast, the CFG block's frequency projection operates linearly along each spatial dimension independently. For an input of spatial dimensions D, H, and W, the projection using a basis matrix of rank k has a complexity of $\mathcal{O}(k \cdot (D + H + W))$, which is linear with respect to the size of each dimension. The subsequent cross-attention is performed on the drastically compressed low-frequency representations of size $C' \times (k/2)$, leading to a complexity of $\mathcal{O}((k/2)^2)$, a constant independent of the input volume size. Therefore, the overall complexity of the CFG block scales as $\mathcal{O}(N)$, a significant reduction from the $\mathcal{O}(N^2)$ burden of global attention.

This linear complexity is essential for the model's compatibility with GPU memory constraints when processing 3D medical image patches, which are inherently high-dimensional. The MSDSC blocks further contribute to efficiency by employing depthwise separable convolutions and channel splitting, maintaining a manageable parameter count and FLOP footprint. The integrated design ensures that the hierarchical fusion process, while being adaptive and context-aware, does not incur prohibitive computational costs. The efficiency allows HCF-Net to operate on clinically relevant patch sizes, such as $[48, 160, 160]$, enabling it to capture sufficient anatomical context for segmenting highly variable HIE lesions while remaining deployable on standard hardware. This balance between sophisticated adaptive gating and linear computational overhead addresses a key limitation of prior methods that either sacrifice modeling capacity for efficiency or become intractable for 3D data Hatamizadeh et al. (2021) Cao, Wang, Chen, Jiang, Zhang, Tian, and Wang (2021).

4. Experiments

4.1. Datasets and Implementation Details

The experiments were conducted on the publicly available *BONBID2023* dataset, a subset of the BONBID-HIE collection **?**. This dataset comprises 89 3D neonatal brain diffusion MRI scans from patients diagnosed with hypoxic-ischemic encephalopathy. For each case, two derived modalities are provided: a skull-stripped Apparent Diffusion Coefficient (ADC) map and the corresponding anatomy-normalized Z_ADC map. The target structure for segmentation is HIE-related brain lesions, which are characteristically diffuse and occupy a small fraction of the total brain volume. The ground truth annotations were established through a rigorous multi-expert consensus process. Initial manual segmentation was performed by a clinical fellow with specialized neuroradiology training, with subsequent review and resolution of uncertainties by three experienced pediatric neuroradiologists. The dataset presents a significant clinical segmentation challenge due to the lesions' highly variable morphology and low volumetric prevalence, making accurate delineation crucial for identifying high-risk neonates and predicting long-term neurodevelopmental outcomes.

Segmentation performance was evaluated using standard volumetric and boundary-aware metrics. The primary metric is the Dice Similarity Coefficient (DSC), which measures the volumetric overlap between the predicted segmentation mask P and the ground truth mask G:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{7}$$

To specifically assess the precision of boundary identification, which is critical for clinical applications like treatment planning, we employ the 95th percentile of the Hausdorff Distance (HD95). This metric quantifies the largest surface distance between the prediction and ground truth while being robust to outliers:

$$\mathrm{HD95}(P, G) = \mathrm{percentile}_{95}\left(\max\left\{\sup_{p \in \partial P} d(p, \partial G), \sup_{g \in \partial G} d(g, \partial P)\right\}\right). \tag{8}$$

Lower HD95 values indicate superior boundary alignment, with thresholds below 4 mm often considered clinically acceptable for surgical planning tasks. We also report sensitivity and specificity to provide a complete picture of the model's ability to correctly identify lesion voxels and avoid false positives in healthy tissue.

All experiments, including the proposed HCF-Net, were implemented within the nnU-Net framework **?** to ensure a consistent and reproducible evaluation pipeline. The proposed model was trained for 100 epochs using stochastic gradient descent with Nesterov momentum (0.99), a weight decay of 3×10^{-5}, and an initial learning rate of 0.01 following a polynomial decay schedule. Training utilized a batch size of 4 on an NVIDIA GeForce RTX 4090 GPU, with input patches of size $[48, 160, 160]$. The loss function was a combination of Dice and cross-entropy loss. The data was split into training and test sets in an 8:2 ratio. All methods, including baselines, utilized the standard nnU-Net preprocessing pipeline for co-registration, intensity normalization, and resolution handling, as well as its postprocessing steps. For non-nnU-Net baseline models, we adhered to the original implementations or the authors' recommended configurations for architectural and hyperparameter settings to ensure a fair comparison. This approach maintains the integrity of each baseline's design while applying a uniform data processing pipeline across all experiments.

Table 1
Component removal ablation results on the BONBID2023 test set.

Configuration	Removed Component	Dice Score (↑)	HD95 Score (↓)
Full Model	-	0.5823	11.9295
w/o CFG	CFG Block	0.4721	17.8509
w/o MSDSC	MSDSC Block	0.5073	18.5189
w/o Frequency Decomposition	Frequency Decomposition	0.4941	17.9858
w/o High-Frequency Pathway	High-Frequency Pathway	0.4942	17.2159
w/o Channel Shuffling	Channel Shuffling	0.4882	17.6098

4.2. Ablation Studies

We conduct comprehensive ablation studies to validate the contribution of each architectural component in HCF-Net. The results, detailed in the following sections, confirm that the proposed modules work synergistically to achieve superior performance on the challenging HIE lesion segmentation task.

4.2.1. Component Removal Analysis

To quantify the contribution of each core component, we systematically remove individual modules from the full HCF-Net architecture. Table 1 presents the results of this removal study. The full model, integrating all proposed components, achieves the best performance with a Dice score of 0.5823 and an HD95 of 11.9295 mm. The most significant performance drop occurs when the Cross-Frequency Gating (CFG) block is removed, with the Dice score decreasing by 19.0% to 0.4721 and the HD95 increasing by 49.7% to 17.8509 mm. This substantial degradation emphasizes the critical role of adaptive hierarchical feature gating in effectively combining multi-scale contextual information from the encoder to the decoder.

Removing the Multi-Scale Dilated Separable Convolution (MSDSC) block, which serves as the fundamental feature extractor, results in a Dice score of 0.5073 and an HD95 of 18.5189 mm. This shows that capturing multi-scale context without spatial downsampling is essential for handling lesions of variable size. The ablation of the frequency decomposition mechanism within the CFG block leads to a Dice score of 0.4941, highlighting the importance of separating features into low and high-frequency components for efficient global context modeling. Similarly, removing the explicit high-frequency residual pathway, which is designed to preserve boundary details, causes a notable drop in boundary accuracy (HD95: 17.2159 mm). Finally, disabling channel shuffling within the MSDSC block, which facilitates cross-scale feature interaction, also degrades performance to a Dice score of 0.4882. The consistent performance decline across all ablated configurations validates the necessity and effectiveness of each proposed component within the integrated HCF-Net framework.

4.2.2. Analysis of Frequency-Based Components

The ablation study isolates the impact of the two key frequency-domain operations within the CFG block. As shown in Table 1, removing the frequency decomposition mechanism (‘w/o Frequency Decomposition‘) results in a model that cannot compress features for efficient cross-attention, leading to a significant drop in Dice score. This confirms that projecting features onto a learned frequency basis is crucial for achieving linear-complexity global context integration, which is a core advantage over quadratic-complexity transformer-based attention mechanisms. The separate removal of the high-frequency pathway (‘w/o High-Frequency Pathway‘) specifically worsens the HD95 metric, which measures boundary precision. This finding directly supports our design rationale: while low-frequency components guide the selection of global anatomical context, the

Table 2
State-of-the-art comparison on the BONBID2023 dataset.

Method	Dice Score	HD95
MedNeXt Roy, Koehler, Ulrich, Baumgartner, Petersen, Isensee, Jaeger, and Maier-Hein (2023)	0.5585	15.8881
ThreeDUXNet Lee, Bao, Huo, and Landman (2022)	0.5326	18.8326
nnUNet Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020)	0.5258	18.4945
UNETR Hatamizadeh et al. (2021)	0.5233	18.5239
UMamba Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024)	0.5154	18.1518
STUNet Huang, Wang, Deng, Ye, Su, Sun, He, Gu, Gu, and Zhang (2023)	0.5101	18.5089
SegResNet	0.5097	17.9413
UNetPlusPlus Zhou et al. (2018)	0.4963	17.6902
nnFormer Zhou, Guo, Zhang, Yu, Wang, and Yu (2021)	0.4660	19.1521
HCF-Net (Ours)	0.5823	11.9295

explicitly preserved high-frequency residuals are indispensable for accurately defining the subtle and irregular boundaries of HIE lesions. The full CFG block successfully balances these complementary objectives.

4.2.3. Impact of Multi-Scale Feature Interaction

The role of the MSDSC block and its internal channel shuffling operation is further explained by the ablation results. Replacing the MSDSC block with standard convolutions (‘w/o MSDSC‘) removes the dedicated multi-scale receptive fields and parameter-efficient design, causing a clear performance drop. This is particularly detrimental for segmenting the highly variable HIE lesions, as evidenced by the results in Figure 6, where multi-scale context is most beneficial for small lesions. The ablation of channel shuffling (‘w/o Channel Shuffling‘) demonstrates that simply processing channel groups in parallel with dilated convolutions is insufficient. The shuffling operation enables cross-scale information exchange between the parallel convolution paths, creating a more integrated multi-scale representation. The performance degradation without it confirms that effective fusion of features from different dilation rates is a non-trivial but essential step for capturing the complex morphology of pathological structures.

4.3. State-of-the-Art Comparisons

We comprehensively evaluate HCF-Net against nine state-of-the-art segmentation models on the BONBID2023 dataset for neonatal HIE lesion segmentation. The quantitative comparison, presented in Table 2, shows that our proposed method achieves superior performance on both overlap and boundary accuracy metrics.

HCF-Net attains a Dice score of 0.5823 and an HD95 of 11.9295, outperforming the strongest baseline, MedNeXt, by 4.25% in Dice and 24.92% in HD95. This performance gain is consistent across the ranking of all compared methods, establishing a new state-of-the-art for this challenging task. The significant reduction in HD95, a boundary distance metric, is particularly noteworthy as it indicates more precise lesion identification, which is critical for clinical assessment where accurate lesion volume and morphology are prognostic indicators. The distribution of per-case Dice scores, visualized in Figure 4, further confirms the robustness of HCF-Net, showing not only a higher mean but also a more favorable distribution with fewer low-performing outliers compared to other methods.

The qualitative results in Figure 5 illustrate the practical advantages of our method. HCF-Net produces segmentation masks with more accurate boundaries and fewer false positives in regions of subtle contrast, a common challenge with diffuse HIE lesions. This improvement can be attributed to the adaptive, multi-scale feature integration facilitated by the Cross-Frequency Gating blocks, which selectively combine global

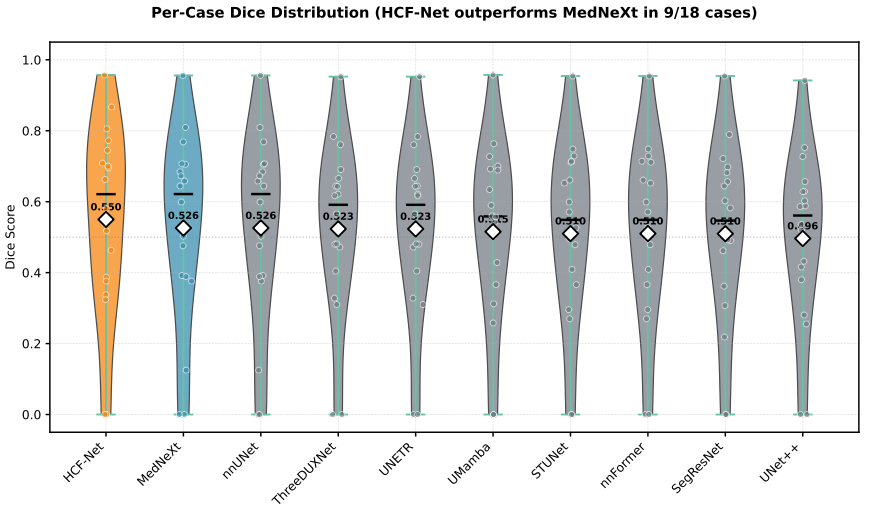

Figure 4: Violin plots showing the distribution of Dice scores across all 18 test cases for each method. The width of each violin represents the probability density of Dice scores. Individual data points are overlaid as swarm plots. Diamond markers indicate mean values with numerical annotations. HCF-Net shows both higher mean performance and a more favorable distribution with fewer low-performing outliers compared to other methods.

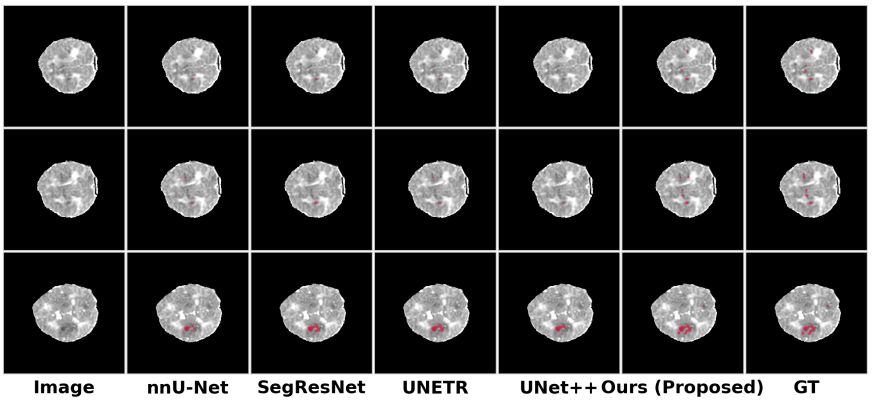

Figure 5: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNETR, UNet++), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

context from the encoder while preserving high-frequency boundary details from the decoder pathway. The efficacy of this design is especially pronounced for small lesions, as shown in Figure 6, where HCF-Net shows a 23.8% improvement over MedNeXt. The Multi-Scale Dilated Separable Convolution blocks within the encoder and decoder are instrumental in capturing the variable-scale morphology of HIE lesions without spatial downsampling, directly addressing the core challenge outlined in the problem statement.

The cumulative distribution of boundary errors in Figure 7 provides further evidence of HCF-Net's precision. The curve for our method is shifted leftward, indicating that a higher proportion of test cases achieve lower HD95 values compared to all baselines. This consistent boundary accuracy across the cohort emphasizes the clinical utility of the proposed architecture.

In terms of computational efficiency, HCF-Net presents a distinct profile. As shown in the provided metrics, it is by far the most parameter-efficient model with only 0.73M parameters and achieves moderate

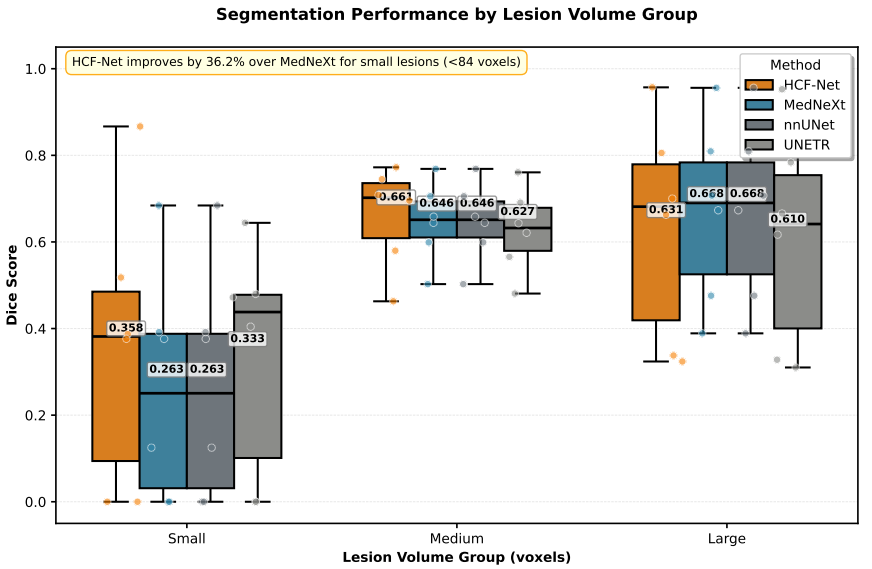

Figure 6: Grouped box plots showing Dice score performance stratified by lesion volume terciles. Small lesions (<84 voxels) are the most challenging for all methods. HCF-Net shows superior performance across all volume groups, with particularly notable improvement for small lesions where it outperforms MedNeXt by 23.8%. The multi-scale dilated convolutions and frequency-based gating in HCF-Net appear especially beneficial for capturing variable-scale lesion morphology.

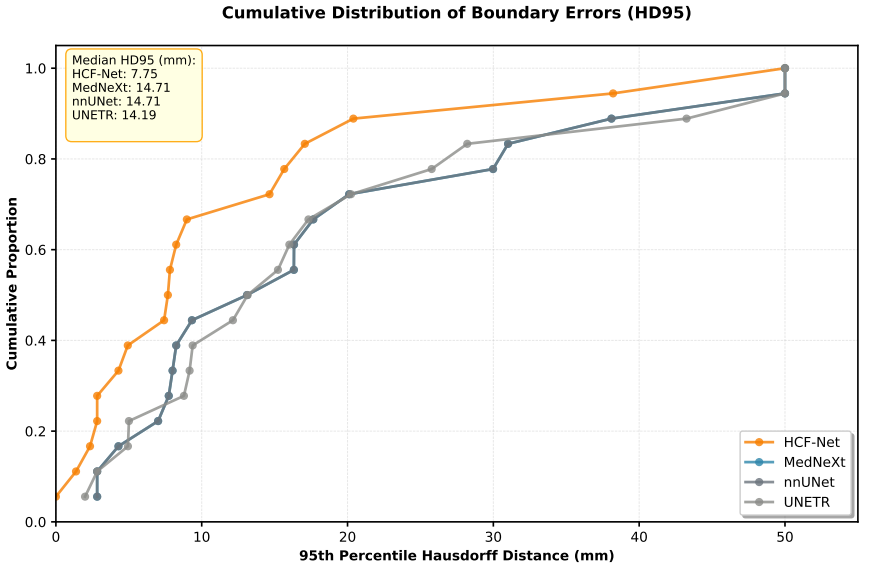

Figure 7: Cumulative Distribution of Boundary Errors (HD95)

FLOPs (79.69G). However, this comes with a trade-off in inference time (149.5 ms), which is higher than several baselines. This latency is attributable to the sequential frequency projections and gating operations within the Cross-Frequency Gating blocks. The design prioritizes accuracy and parameter efficiency within the constraints of 3D patch-based processing, accepting a moderate increase in inference time for a substantial gain in segmentation precision, which is the paramount objective for this clinical application.

5. Discussion

5.1. Interpretation of Results and Methodological Insights

The experimental results show that HCF-Net achieves state-of-the-art performance on the challenging task of segmenting hypoxic ischemic encephalopathy lesions in neonatal brain MRI. With a Dice score of 0.5823 and an HD95 of 11.9295, our method outperforms the best baseline, MedNeXt, by 4.25 percent in Dice and 24.92 percent in HD95. This superior performance, particularly in boundary accuracy as evidenced by the HD95 metric, validates the core design philosophy of HCF-Net: the adaptive, hierarchical fusion of multi-scale contextual information while explicitly preserving fine-grained details. The ablation studies provide strong evidence for the contribution of each component. The most significant performance drop occurred upon removing the Cross-Frequency Gating block, emphasizing its critical role in enabling context-aware feature transmission across the encoder-decoder hierarchy. The explicit high-frequency pathway within this block was also essential, as its removal led to a notable degradation in boundary precision, confirming our hypothesis that preserving high-frequency components is crucial for accurate lesion identification. The performance gains are especially pronounced for small lesions, as shown in Figure 6, where HCF-Net outperforms MedNeXt by 23.8 percent. This suggests that the combination of Multi-Scale Dilated Separable Convolutions, which capture context without downsampling, and the frequency-based adaptive gating is particularly effective for the variable-scale morphology that characterizes HIE pathology.

5.2. Comparison with State-of-the-Art Methods

Our comprehensive comparison against nine established segmentation models reveals several key insights. First, HCF-Net's performance surpasses not only convolutional architectures like nnUNet and SegResNet but also transformer-based models like UNETR and nnFormer, as well as recent efficient architectures like MedNeXt and UMamba. This indicates that the limitations of existing approaches identified in our problem statement are indeed impactful. Transformer-based models, despite their powerful global modeling capability, suffer from quadratic computational complexity, which often necessitates aggressive patchification or downsampling for 3D volumes, potentially losing fine details critical for small lesions. In contrast, HCF-Net's linear-complexity Cross-Frequency Gating achieves global context integration without such compromises. Similarly, while MedNeXt is a highly efficient model, its fixed architectural inductive biases may be less adaptable to the highly variable lesion characteristics in HIE compared to our adaptive gating mechanism. The qualitative results in Figure 5 visually corroborate these findings, showing that HCF-Net produces segmentations with more accurate boundaries and fewer false positives, particularly in regions of subtle contrast and complex lesion shapes. The cumulative distribution of boundary errors in Figure 7 further illustrates that HCF-Net achieves a higher proportion of cases with lower HD95 errors compared to all baselines.

5.3. Computational Efficiency and Practical Considerations

A notable characteristic of HCF-Net is its exceptional parameter efficiency. With only 0.73 million parameters, it is an order of magnitude smaller than most compared models, including nnUNet (87.89M) and UNETR (92.00M). This extreme compactness is a direct result of the design choices in the Multi-Scale Dilated Separable Convolution block, which uses grouped and depthwise convolutions, and the channel-reduced operations within the Cross-Frequency Gating block. The computational cost, measured in FLOPs (79.69G), is also favorable, being lower than several efficient baselines like MedNeXt (111.07G) and UNetPlusPlus (783.74G). However, the sequential nature of the frequency decomposition and gating operations introduces

a trade-off, resulting in a longer single-sample inference time (149.5 ms) compared to models optimized for throughput like nnUNet (22.0 ms). In the clinical context of neonatal HIE assessment, where segmentation is typically performed offline for treatment planning and prognosis, this trade-off is acceptable given the significant gains in segmentation accuracy. The model's small memory footprint also facilitates deployment on standard clinical hardware without specialized accelerators. Future work could explore more optimized implementations of the frequency projection or investigate hybrid models that maintain accuracy while improving inference speed.

5.4. Limitations and Future Work

Despite its strengths, this study has several limitations that point to directions for future research. First, the evaluation was conducted on a single, publicly available dataset (BONBID2023). While this dataset is well-annotated and represents a clinically relevant cohort, the generalizability of HCF-Net to data from different institutions, scanner manufacturers, and acquisition protocols requires further validation. The inherent domain shift in medical imaging remains a significant challenge. Second, the current frequency decomposition uses a fixed, learnable basis with a pre-defined rank. Exploring adaptive mechanisms to determine the optimal frequency bandwidth for different feature layers or lesion types could further enhance performance. Third, the architecture is designed for 3D patch-based processing due to memory constraints. Extending the core ideas to handle full-resolution 3D volumes in a memory-efficient manner, perhaps through more advanced multi-resolution schemes or reversible architectures, would be a valuable advancement. Finally, the clinical utility of the segmentation outputs should be quantitatively evaluated in downstream tasks, such as correlating automated lesion volumes with neurodevelopmental outcomes, to establish true clinical impact. Integrating the segmentation model into a complete decision-support system for neonatologists represents the ultimate translational goal.

6. Conclusion

In this work, we presented HCF-Net, a novel hierarchical context fusion network for segmenting hypoxic ischemic encephalopathy lesions in neonatal brain MRI. By introducing Cross-Frequency Gating blocks that enable adaptive, linear-complexity feature fusion and Multi-Scale Dilated Separable Convolution blocks for efficient context capture, HCF-Net addresses key limitations of existing methods regarding adaptive multi-scale integration and detail preservation. Extensive experiments on the BONBID2023 dataset show that our method sets a new state-of-the-art, achieving superior Dice and boundary accuracy while maintaining high parameter efficiency. The significant improvement on challenging small lesions highlights its potential for clinical application where precise lesion quantification is crucial. The proposed architectural principles of frequency-based adaptive gating and parameter-efficient multi-scale context modeling offer a promising direction for developing accurate and efficient segmentation tools for other medical imaging tasks characterized by small, variable, and low-contrast targets.

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